JOINT ESTIMATION OF PIANO DYNAMICS AND METRICAL STRUCTURE WITH A MULTITASK MULTI-SCALE CNN

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ABSTRACT

Estimating piano dynamic from audio recordings is a fundamental challenge in computational music analysis. In this paper, we propose an efficient multitask CNN that jointly predicts dynamic levels, change points, beats, and downbeats from a shared latent representation. These four targets form the metrical structure of dynamics in the music score. Inspired by recent vocal dynamic research, we use the multi-scale CNN backbone to leverage Bark-scale specific loudness as input features. Compared to log-Mel as input, this reduces model size from 13.4 M to 0.5 M, enabling long audio sequential input. We indentify the 60-second as optimial for combined tasks, which doubled the length of beat tracking commonly used. Evaluated on the public MazurkaBL dataset, our model achieves state-of-the-art results across all tasks. This work sets a new benchmark for piano dynamic estimation and delivers a powerful and compact tool, paving the way for large-scale, resource-efficient analysis of musical expression.

***Index Terms*—** Piano dynamic estimation, Beat and downbeat tracking, Bark-scale specific loudness, Multi-scale network, Multitask learning.

1. INTRODUCTION

The creation, comprehension and reproduction of music are fundamental aspects of human culture. In western musical tradition, dynamics-the variations in loudness-is a key aspect of musical expression [1]. Indicated in a score by markings such as *p* (*piano*, soft) and *f* (*forte*, loud), dynamics are essential for shaping musical phrases, conveying emotion, and articulating structural differentation. This vocabulary extended to a nuanced of static levels, from *pp* (*pianissimo*) to *ff* (*fortissimo*), and includes gradual transitions like *crescendo* and *decrescendo*. The computational modeling of these expressive markings is valuable for music education, performance analysis, and theory-informed music generation [2].

The core challenge in estimation from audio lies in the inherent relativity and ambiguity of dynamic markings. A symbol like *pianissimo* does not map to a fixed physical measurement like a decibel level. Instead, its interpretation is deeply contextual, influenced by musical style, performer's intent, and the acoustic environment [3,4]. This lack of a standardized ground truth has historically posed a significant challenge for machine learning algorithms, often leading to poor generalization among different performers or pieces [5, 6].

To circumvent the ambiguity of music dynamics, researchers often adopt proxy targets: LUFS in audio generation and MIDI velocity in music transcription and analysis [7–9]. However, MIDI velocity reflects performer's physical action rather than the perceived loudness, and is entangled with music instrumental timbre and touch [10]. Although automatic music transcription (AMT) systems can accurately estimate MIDI velocity from Yamaha Disklavier piano performance, estimate across diverse piano remains unsolved [11]. The subsequent task of regressing dynamic markings from MIDI velocity expectionally difficult.

Given these complexities, we propose an end-to-end multitask learning approach to estimate dynamic markings and their underlying metrical structure directly from audio. Inspired by recent advances in vocal dynamics estimation [12], we use Bark-scale specific loudness (BSSL) as input feature. This representation is processed by a multi-scale attention CNN backbone, adapted from [13], to extract a shared latent representation. This unified representation is then channeled through a Multi-gate Mixture-of-Experts (MMoE) layer [14], which generates specialized features for four distinct task heads that jointly predict: (1) dynamic levels, (2) change points, (3) beat positions, and (4) downbeat positions. Together, these targets model not only the dynamic markings but also their underlying metrical structure, as the beat and downbeat grid forms the rhythmic foundation of a musical score.

The proposed multitask framework offers significant practical applications compared to single-task approaches. Its primary use is to enrich a musical score with dynamic markings, which is particularly valuable for enhancing the output of score-level automatic music transcription systems that typically capture only musical notes. This is achieved by combining the model's predicted dynamic levels and change points and aligning them to a beat grid (see Fig. 1). A more advanced application stems from the model's end-to-end design, where it can predict the beat and downbeat grid by itself, using these internal predictions to post-process the dynamics. This enables fully audio-based music analysis where no ground-truth score or annotations are available. By achieving state-of-the-art results on the expressive MazurkaBL dataset, we present this multitask CNN as a complete, end-to-end tool that supports large-scale music performance analysis, making it a valuable resource for the music research community.[1]

2. METHODOLOGY

2.1. Bark-scale Feature Extraction (WILL BE TIDY UP)

The Bark-scale specific loudness (BSSL) feature, based on a psychoacoustic model of perceived loudness over critical bands, has proven more effective than common log-Mel spectrograms for tasks like vocal dynamics estimation [12]. However, the implementation in that study relied on the MOSQITO toolbox [15], which is designed for robust sound quality assessment and thus mandates a 48 kHz sampling rate with constrained FFT parameters. Since music

[1]Code and pre-trained models are available at: https://github.com/zhanhe/piano-dynamic-estimation

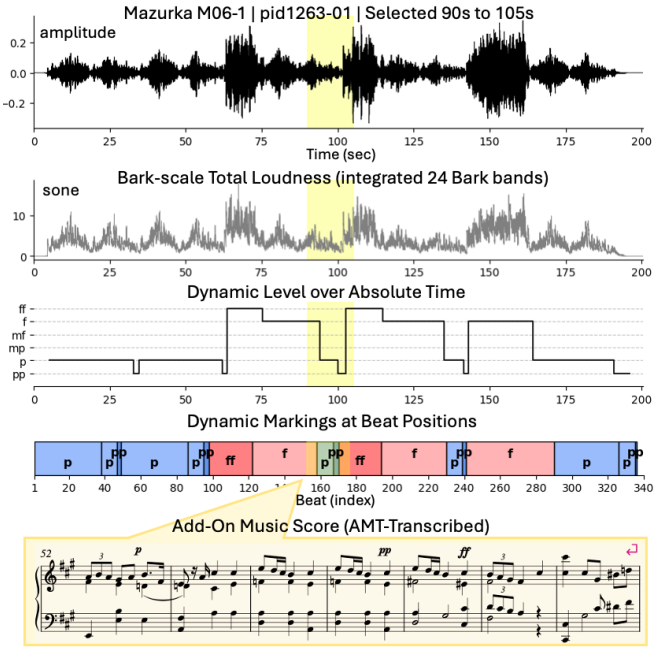


Fig. 1. Workflow for adding dynamic markings to an AMT-transcribed score from audio. The highlighted region (90–105 s of Chopin Mazurka Op. 6 No. 1) shows a case study. Dynamic markings is in music beat grid, with the meaning of dynamic levels at its shifting location (i.e., the change point).

recordings are mostly encoded at 44.1 kHz, using MOSQITO would necessitate upsampling, a computationally expensive process that can introduce artifacts. We developed a more flexible feature extractor in PyTorch framework, which reproduces the `ma_sone` function from Pampalk's widely-used music analysis MATLAB toolbox [16], following the processing chain proposed in [17, 18].

The feature extraction pipeline proceeds as follows. First, the audio is converted to mono, peak-normalized to -1 dBFS, and resampled to $f_s = 22.05$ kHz. To ensure comparability with the recent SOTA beat tracking model [19], we use identical STFT parameters: a Hann window $w[n]$ of length $N = 1024$ and a frame rate of FPS $= 50$ (i.e., 20 ms per frame and hop size of 441), where n indexes samples within a frame. The STFT yields the complex spectrogram $X(k,t)$, with frame index t and frequency bin index k. We then form the power spectrogram $P(k,t)$ with window-energy normalization, and subsequently convert to decibel scale:

$$P_(k,t) = \frac{|X(k,t)|^2}{(\sum_{n=0}^{N-1} w[n])^2}, \tag{1}$$

$$P_{\mathrm{dB}}(k,t) = \max(0, 10\log_{10}(P(k,t))). \tag{2}$$

To approximate the outer and middle-ear frequency response, we apply the Terhardt weighting function $W_{\mathrm{dB}}(f_k)$ [20] designed in the decibel domain, where f_k denotes the center frequency (in kHz) of the k-th bin:

$$W_{\mathrm{dB}}(f_k) = -3.64\, f_k^{-0.8} + 6.5\, e^{-0.6\,(f_k-3.3)^2} - 10^{-3}\, f_k^4, \tag{3}$$

This weighting must be converted to a linear power weight before being applied to $P(k,t)$:

$$W(f_k) = 10^{\,W_{\mathrm{dB}}(f_k)/20}\,; \quad P_W(k,t) = P(k,t)\cdot W(f_k)^2. \tag{4}$$

Next, the weighted power spectrogram $P_W(k,t)$ is mapped to the Bark scale by summing energies up to 24 critical bands [21]. A total of 22 Bark bands are used, which is consistent with our Nyquist frequency of 11.05 kHz, where the 22nd band extends up to 9.5 kHz. Denoting the band index as $b \in \{1,\ldots,22\}$, the Bark spectrogram is:

$$B(b,t) = \sum_{k\in \mathrm{Band}b} P_W(k,t). \tag{5}$$

To account for spectral masking, where a loud sound makes weaker sounds at nearby frequencies harder to preceive, we apply the spreading function following Schoroeder et al. [22]:

$$G(\Delta b) = 15.81 + 7.5(\Delta b + 0.474) - 17.5\sqrt{1+(\Delta+0.474)^2}, \tag{6}$$

with $\Delta b = b_j - b_i$ is the distance between the j-th and i-th bark bands.

S as matrix multiplication :

$$B_S(b,t) = \sum_{i=1}^{22} S_{bi}\cdot B(i,t), \tag{7}$$

where the element S_{bi} of the spreading matrix S is defined by the distance between the j-th and i-th Bark bands. Letting $\Delta z = b - i + 0.474$,

$$S_{bi} = 10^{(15.81+7.5\Delta z - 17.5\sqrt{1+(\Delta z+0.474)^2})/10}. \tag{8}$$

Finally, the masked Bark spectrogram $B_S(b,t)$ is converted to the decibels (dB), and then to the perceived loudness in sones via a standard piecewise [23]:

$$B_{\mathrm{dB}}(b,t) = \max(0, 10\log_{10}(B_S(b,t))). \tag{9}$$

$$B_{\mathrm{sone}}(b,t) = \begin{cases} 2^{(B_{\mathrm{dB}}(b,t)-40)/10}, & B_{\mathrm{dB}}(b,t) \geq 40 \\ (B_{\mathrm{dB}}(b,t)/40)^{2.642}, & otherwise. \end{cases} \tag{10}$$

The resulting BSSL matrix $B_{\mathrm{sone}}(b,t)$ is a 22-dimensional time-frequency representation of perceived loudness. While its aggregated total loudness is useful for visualization (see Fig. 1), we use the complete 24-dimensional matrix as input feature to preserve the vital spectral envelope information.

2.2. Model Architecture

The proposed model is illustrated in Fig. 2. With divergent acoustic requirements, dynamic estimation benefits from a large temporal receptive field [12], whereas beat and downbeat tracking require high temporal resolution to locate transient onsets [24]. To address this need for varied receptive fields, we adapt the multi-scale CNN from [13] as a shared encoder. Given an $F\times T$ time–frequency feature as input, the encoder operates on three parallel streams at different temporal resolutions, corresponding to lengths of T, T/s, and T/s^2. The downsampling and upsampling by the scaling factor s are implemented via setting strides in max-pooling and transposed convolutions. Unlike the original design [13], which used a fix factor of 3, we treat the scaling factors as hyperparameters and study task-dependencies via ablations. We also replace the original task-specific classifiers with a shared latent sequence $\mathbf{Z} \in \mathbb{R}^{T\times 8}$ consumed by all tasks. The model employs a compact configuration with a shared latent dimension of 8, a temporal scaling factor of 5, and an MMoE block with 8 experts.

To specialize the shared representation and mitigate negative transfer between the acoustically heterogeneous tasks (multi-class

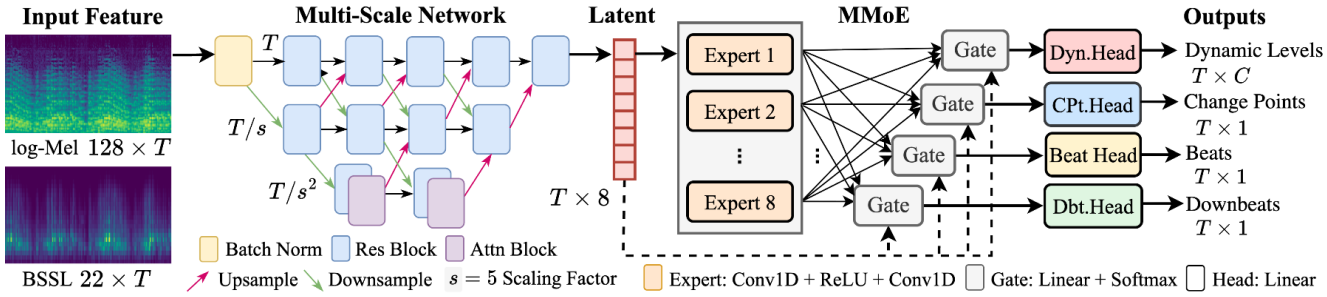


Fig. 2. Proposed model architecture. We apply an empirically optimal temporal scaling factor of 5, and predict dynamic levels across 6 classes: a *blank* class for silence before the first annotation, and the five dynamic classes (*pp, p, mf, f, ff*) from the MazurkaBL dataset.

dynamics vs. other binary targets), we employ the MMoE module as a task-aware decoder [14]. This module consists of a pool of 8 shared experts (a sequential Conv1D blocks) and four task-specific gates. For each task and time steps, a linear gate maps the shared latent feature to expert logits and normalizes them into non-negative weights that sum to one across experts by applying the softmax function; the task feature is the weighted sum of expert outputs under these weights. Finally, separate linear heads map these task-specific features to the output logits, with shapes of $C \times T$ for dynamic level, and $1 \times T$ for the beat, downbeat, and change point.

2.3. Task-specific post-processing

The model's raw output is a sequence of frame-wise probabilities, which we convert into discrete musical events via tailored post-processing. For beat and downbeat detection, we follow [19], identifying events by thresholding probabilities at 50% and applying peak-picking within a 70 ms neighborhood (± 3 frames at 50 fps).

Our final target, a set of discrete dynamic markings, is obtained by sampling the continuous dynamic level (a curve) at the precise location of each change point. We detect these two intermediate components from the model's output: the dynamic level is determined by an $\arg\max$ over class probabilities, and change points are selected by peak-picking on probabilities exceeding a 75% threshold.

An additional post-processing step is applied to change points. While it is not a strict musical rule that dynamic changes occur exactly on beat positions (they can occur between beats), the annotations in the MazurkaBL dataset are exclusively at beat locations. To be consistent with prior work [25], we snap the detected change points to the nearest beat position to ensure a comparable evaluation.

2.4. Loss Function

Training the multitask model involves a composite loss that combines specialized objectives for each task. Our multitask loss, $\mathcal{L}_{\text{MTL}}$, is defined as the sum of four task-specific losses:

$$\mathcal{L}_{\text{MTL}} = \mathcal{L}_{\text{Dyn}} + \mathcal{L}_{\text{CPt}} + \mathcal{L}_{\text{Bt}} + \mathcal{L}_{\text{Dbt}} \tag{11}$$

where each component is defined as follows. For the binary targets (change points, beats, and downbeats), the loss terms $\mathcal{L}_{CPt}$, $\mathcal{L}_{Bt}$, and $\mathcal{L}_{Dbt}$ employ the shift-tolerant weighted binary cross-entropy proposed in [19]. This function addresses two key challenges: it counteracts the extreme sparsity of targets by weighting positive frames more heavily, and it accommodates annotation timing imprecision by incorporating a ± 3 frame tolerance window. For the multiclass target, dynamics, the loss term $\mathcal{L}_{Dyn}$ is a standard cross-entropy masked by the ground-truth beat positions. The masking operation enforces the prelimiary knowledge that dynamic changes occur at beat positions, guiding the model to ignore spurious inter-beat fluctuations.

3. EXPERIMENTS

3.1. Dataset

This experiment uses the MazurkaBL dataset [6], a score-aligned corpus of 2,098 solo piano recordings covering 44 Chopin Mazurkas. After excluding two opuses with irregular dynamics annotations, 1,999 recordings remained. MazurkaBL is the largest resource for studying notated dynamics versus performed loudness from audio, providing score-aligned beat times and verified expressive markings, with a specific focus on dynamics. In contrast, other large datasets like MAESTRO [26] use MIDI velocity as a proxy for dynamics, while others such as ASAP [27] provide score-aligned annotations but lack dynamic markings.

3.2. Implementation details

We use a 5-fold cross-validation protocol, stratified by Mazurka opus, to train and evaluate our model. Within each fold, the model is trained for 120 epochs, with the best-performing checkpoint selected based on the F1 score on its respective validation set. The training configuration includes the AdamW optimizer with a learning rate of 3e-4, a batch size of 10, and a fixed random seed of 86. For data augmentation, we slice audio into 60-second segments with 50% overlap during training, while no overlap is used for evaluation. Both BSSL (22 bark bands) and log-Mel (128 mel bins) features are extracted using the same STFT parameters as in [19] to ensure identical temporal resolution. When run on a NVIDIA RTX 3090 (24 GiB), a full 5-fold run completed in approximately 20 hours, with peak GPU memory usage of 4 GiB.

3.3. Baselines

We compare the proposed multitask CNN against the single-task CNNs and task-specific baselines.

Single-task CNN. The single-task CNN baselines are adapted from the public implementation of the multi-scale CNN [13], training one independent model per target. To ensure a fair comparison, the training setup mirrors our multitask model, and the loss is the correspond-

Table 1. Basic evaluation results on different tasks, reported F1 scores as mean ± standard deviations (%) over 5 folds cross-validation. BS and PELT are algorithm-based methods with no trainable parameters, while the ANN did not report this attribute.

Method	Feature	Dynamic Level F1	Change Point F1	Beat F1	Downbeat F1	# Params
ANN [25]	$\text{BSSL}_{\text{sone}}$	29.4	–	–	–	n/a.
PELT [25]	$\text{BSSL}_{\text{sone}}$	–	10.8	–	–	n/a.
TCN+DBN [24]	log-Mel	–	–	60.9 ± 1.8	30.4 ± 1.3	0.3 M
Beat This [19]	log-Mel	–	–	80.5 ± 2.7	52.8 ± 6.4	20.3 M
Single-task CNN [13]	$\text{BSSL}_{\text{sone}}$	47.7 ± 7.5	16.8 ± 5.0	83.7 ± 1.4	44.1 ± 0.9	0.4 M
w/o. BSSL	log-Mel	46.4 ± 7.9	16.6 ± 2.4	83.1 ± 1.4	48.1± no yet	13.3 M
Multitask CNN (Proposed)	$\text{BSSL}_{\text{sone}}$	51.4 ± 5.4	21.6 ± 5.9	83.4 ± 0.8	52.3 ± 1.2	0.5 M
w/o. BSSL	log-Mel	48.5 ± 6.5	19.8 ± 7.3	83.7 ± 1.6	56.1 ± 2.9	13.4 M
w/o. MMoE	$\text{BSSL}_{\text{sone}}$	48.3 ± 2.6	21.7 ± 5.5	83.9 ± 0.8	53.1 ± 1.6	0.4 M
w/o. temporal scaling (s set to 1)	$\text{BSSL}_{\text{sone}}$	49.6 ± 9.0	13.1 ± 3.3	79.6 ± 1.1	40.7 ± 1.1	0.4 M

ing single term from our multitask loss, with an empirically optimal temporal scaling factor of 5.

Dynamic and change-point baselines. We report published results from representative methods in [25]. This includes the Artificial Neural Networks (ANN) for dynamics, and the Pruned Exact Linear Time (PELT) algorithms for change points. These methods are tuning-intensive and non–end-to-end, so we report the literature scores rather than re-implement them.

Beat and downbeat baselines. We include the time-convolutional network (TCN) with a dynamic Bayesian network (DBN) post-processor [24], and the recent state-of-the-art transformer model, Beat This [19]. Both models are retrained from scratch on the MazurkaBL dataset using their public code and the same 5-fold protocol as ours.

3.4. Evaluation Metrics

We evaluate four tasks using F1 scores. For beat and downbeat tracking, we report F1 with a ± 70 ms tolerance, consistent with prior work [19, 24]. For dynamic levels and change points, we devise two distinct evaluation scenarios.

Basic. This scenario simulates an ideal case where accurate beat information is available, such as refinement to a music score that has reliable beat positions but lacks dynamic markings. Consistent with single-task benchmarks [25], we align the model's predictions with the ground-truth beats. The continuous dynamic levels are sampled at these ground-truth positions, discretized, and then evaluated using a macro-averaged F1 score across five classes (*pp, p, mf, f, ff*), the sixth class *blank* is masked). Change points are snapped to the nearest ground-truth beat and are thus represented on the beat grid for evaluation with a standard F1 score.

Advanced. This scenario assesses the model's practical, end-to-end performance where ground-truth beats are unavailable, such as in audio-only music analysis. The evaluation relies on the model's own predicted beat positions. We first sample the predicted dynamic level curve at these predicted beats to obtain a beat-wise sequence of dynamics. Then, we achor the predicted change points to this beat grid. The final "dynamic markings" are formed by using these change point indices to define the dynamic level for each segment. This output is evaluated using accuracy (Acc) and the micro-averaged F1 score across five dynamic classes .

4. RESULTS

The proposed multitask CNN model demonstrates superior performance across all four tasks in Table 1. It achieves new SOTA F1 scores for both dynamics estimation (51.4%) and change point detection (21.6%). Although a small gap remains in beat tracking when compared with the specialized system "Beat This", the core advantage of our approach is confirmed by the comparison with single-task CNNs. With BSSL features, the multitask CNN achieves improvements in dynamics (+3.7%), change points (+4.8%), and downbeats (+8.2%), with only a minor trade-off in beat detection (-0.3%). Notably, this overall performance is achieved within a single, highly parameter-efficient multitask model that further confirms its practical benefits, as it uses its own predicted beat positions for post-processing rather than relying on ground-truth annotations.

Beyond the performance gains, our analysis reveals a strong task-feature dependency, with BSSL features proving superior for dynamics and log-Mel features excelling at beat tracking. The key advantage of BSSL is its compactness (22 Bark bins in our FFT settings, compared to 128 Mel bins), which dramatically reduces the model size from 13.5 M to 0.4 M trainable parameters. This smaller footprint enables the processing of longer audio sequences, directly benefiting tasks that require long-term temporal information and revealing BSSL's potential for a wider range of musical tasks with long-temporal dependencies.

Furthermore, our ablation study (in Table 1) investigates the impact of the MMoE module and the multi-scale functionality. Removing the MMoE module significantly degrades dynamics estimation while yielding minor gains in other tasks, highlighting its role in optimally allocating weights by light-weight shared experts, instead of a huge structural model changes. Disabling the multi-scale functionality (by setting the scaling factor $s = 1$) impairs the model's ability to perceive features at different temporal resolutions, leading to a performance drop across all tasks. This degradation was particularly severe in change point and downbeat detection, which might requiring high temporal precision.

New: Advance evaluation. Now a figure will replace the **Table2**, to show model actual performance when using model own predicted beat grid – a practical case that "music score" is unknown. We will discuss on this figure. **PENDING THIS PART.**

Table 2. This table is being replaced by a figure to conduct the "Advanced" evaluation.

Dyn. Acc.	Dyn. F1.	Num of CPt_{Pred}	Num of CPt_{GT}
49.5	16.1	78.9	39.0
49.5	16.1	78.9	39.0
49.5	16.1	78.9	39.0

5. CONCLUSION

In this paper, we introduce a compact multitask, multi-scale CNN that jointly estimates piano dynamics, change points, beats, and downbeats directly from audio. Using Bark-scale specific loudness as input and an MMoE decoder, our model leverages long-range temporal context (60 s) while remaining highly parameter-efficient (0.5 M). Evaluated on the MazurkaBL dataset, it achieves SOTA results for dynamics and change point detection and competitive in beat and downbeat tracking. This demonstrates not only the model's practical utility but also its significant potential for broader applications. Future work will focus on combining our proposed model with score-level piano transcription systems. Such a pipeline would produce music scores with expressive dynamics marking more than *pp* to *ff*, and though this presents a new challenge in developing appropriate evaluation methods for such comprehensive outputs.

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